

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**End Semester Examination – Winter 2018****Course: B. Tech in Electrical Engineering****Subject Name: Electrical Engineering Materials****Max Marks: 60****Date: 10/12/2018****Sem: III****Subject Code: BTEEE305A****Duration: 3 Hr.****Instructions to the Students:**

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve following questions.**12**

- A) Define atomic packing factor of the cubic system. Show that the APF for BCC crystal structure is 0.68 **CO 1 4**
- B) With neat diagram explain and derive Bragg's law of X-ray diffraction in crystal. **CO 1 5**
- C) If the interplanar distance for a plane having Miller indices (1 1 0) is $2.86 \text{ \AA}$. Calculate the lattice constant 'a' **CO 1 3**

OR

- C) A certain orthorhombic crystal has a ratio of a:b:c of 0.430:1:0.377. Find the Miller indices of faces whose intercepts are **CO 1 3**
- i) 0.215:1:0.188
- ii) 0.860:1:0.754

Q.2 Solve following questions.**12**

- A) Give the classification with examples for magnetic materials based on relative permeability(μ_r) **CO 2 5**
- B) Draw and explain the hysteresis loop for ferromagnetic materials. What is retentivity and coercivity **CO2 7**

OR

- B) Define magnetic dipole moment & relative permeability(μ_r). Show that $\mu_r = (1 + \chi_m)$, Where χ_m is magnetic susceptibility. **CO 2 7**

Q. 3 Solve following questions.

- A) Describe the formation of energy bands in solids and explain with proper diagrams, classification of materials into conductors and insulators. **CO 3 5**
- B) With ref to superconductivity explain with neat diagrams **CO 3 7**
- i) Meissner effect
- ii) Cooper pair formation
- iii) High temp super conductivity

Q.4 Solve following questions.**12**

- A) Explain Hall effect and its significance. Give its applications **CO 2** **6**
- B) Explain the concept of Fermi-level in case of semiconductors and derive an expression for Fermi-energy in intrinsic semiconductor **CO 3** **6**

OR

- C) Obtain the expression for the
- i) Total conductivity of semiconductor **CO 3** **6**
- ii) conductivity of N & P semiconductors

Q. 5 Solve following questions.**12**

- A) Define Electric dipole moment and explain dielectric polarization and indicate different types of polarizations **CO 4** **6**
- B) Derive Clausius-Mosotti relation from first principles as applied to dielectric materials, state the assumptions made **CO 4** **6**

Q. 6 Solve following questions.**12**

- A) Explain the concept of Nano-technology as it applies to electrical / electronic engineering materials. Also state important properties and applications of Nano particles **CO 4** **6**
- B) With neat diagram explain the principle, construction and working of STM- Scanning Tunneling Microscope **CO 4** **6**

OR

- C) Write a short note on carbon nano-tubes and its applications **CO 4** **6**

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